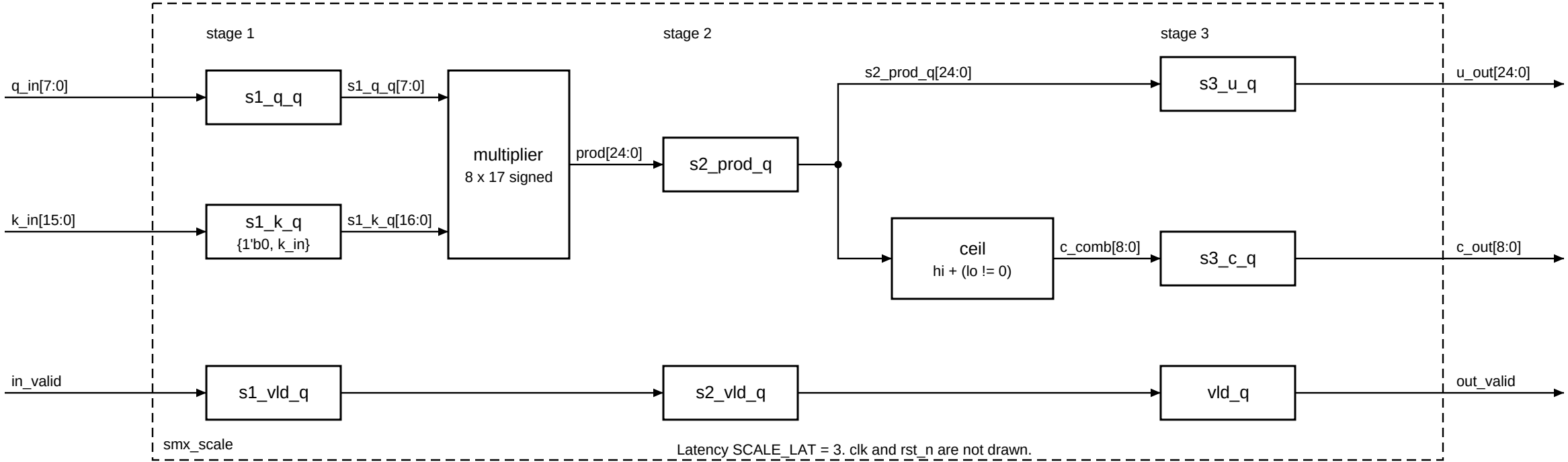
stage 1
stage 2
stage 3
q_in[7:0]
s1_q_q
s1_q_q[7:0]
multiplier
8 x 17 signed
prod[24:0]
s2_prod_q
s2_prod_q[24:0]
s3_u_q
u_out[24:0]
k_in[15:0]
s1_k_q
{1'b0, k_in}
s1_k_q[16:0]
ceil
hi + (lo != 0)
c_comb[8:0]
s3_c_q
c_out[8:0]
in_valid
s1_vld_q
s2_vld_q
vld_q
out_valid
smx_scale
Latency SCALE_LAT = 3. clk and rst_n are not drawn.